

CrashX: Domain Adaptation Removes Omissions but Not Hallucinations in Video-LLM Explanations of Traffic Accidents

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Abstract—A trustworthy explanation of a dashcam collision must name the right vehicles, state the right cause, and place the impact at the right moment. We release CrashX-1500, a dataset of 1,500 dashcam clips for which our annotators wrote the complete forensic ground truth—severity, impact geometry, vehicles by colour and type, weather, crash window, and a multi-sentence causal explanation—and we use it to separate two distinct ways Video-LLMs fail: *omission* (leaving out what happened) and *hallucination* (asserting what did not). Four open Video-LLMs evaluated zero-shot on a 150-video held-out split are fluent but nearly useless: explanation BLEU-4 ≤ 0.026 , and 28–98% of forensic fields are missing. QLoRA adaptation of Qwen2.5-VL-7B on our 1,198 training videos (CrashLogic-7B) raises BLEU-4 to 0.142 and BERTScore from 0.486 to 0.686, and cuts the omission cost from 0.462 to 0.107 ($p < 10^{-4}$, improved on 140 of 150 videos). The hallucination cost, however, does not move ($0.227 \rightarrow 0.223$, $p = 0.81$): one in five adapted explanations names a vehicle whose colour contradicts the ground truth, and the model fabricates a collision in every confirmed no-crash test video. We further show that the adapted model’s apparently strong crash-window localisation (tIoU $0.012 \rightarrow 0.373$) is a learned dataset prior: it emits only two windows, its predicted start is uncorrelated with the true start (Spearman $\rho=0.009$), and a constant window beats it (tIoU 0.408). Finally, we test whether decode-time Temporal Contrastive Decoding (TCD), which subtracts logits obtained from a reversed-frame negative, repairs the residual errors. Under paired tests on 150 videos it does not: mild contrast ($\alpha=0.5$) is harmless but changes no metric significantly, while strong contrast ($\alpha \geq 1.5$) significantly increases both hallucination and omission. We explain why with a simple argument—reversal preserves every order-invariant cue (colour, weather, agent identity), so the contrast cannot touch the errors that dominate—and we release outputs, per-video scores, and an automatic error taxonomy so that future claims about faithful accident explanation can be checked against the same bar.

Index Terms—Traffic accident understanding, Video-LLM, hallucination, omission, explanation faithfulness, temporal grounding, contrastive decoding, QLoRA, Car Crash Dataset

I. INTRODUCTION

Adashcam records a rear-end collision: the camera car closes on a slow black truck and strikes it. Asked to explain the crash, a strong open Video-LLM returns three paragraphs on overcast skies, road surface, and “traffic conditions”. The truck is never mentioned. A domain-adapted version of the same model does name a lead vehicle and a rear-end impact—but calls the truck a red car, and places the impact in a one-second window it uses for more than half of all videos in the test set.

These two outputs fail differently, and the difference matters. The first *omits*: it says nothing false, but nothing useful. The second *hallucinates*: it commits to specific, checkable claims that are wrong. For an insurer, an investigator, or an autonomous-driving safety team, the second failure is worse, because a confident and specific narrative is precisely the kind that gets acted upon [1], [2]. Yet standard captioning metrics reward the second output over the first, and a paper reporting only BLEU or BERTScore would conclude that domain adaptation has “solved” the problem.

a) *This paper.*: We ask a narrower question than “can Video-LLMs explain crashes?”. We ask *which* failures domain adaptation fixes, which it leaves untouched, and whether a popular training-free remedy—decode-time contrastive suppression of temporally implausible tokens [3], [4]—repairs what remains. To study it we build CrashX-1500, our own human-annotated crash explanation resource: 1,500 five-second dashcam clips, each annotated in-house with structured forensic fields (severity, impact location, vehicles by colour and type, weather, camera view, crash window, and an explicit ambiguity note) and a ~ 96 -word free-text explanation, so that every claim a model makes can be scored against a human-written reference (Section III-B).

We evaluate four open Video-LLMs zero-shot, fine-tune Qwen2.5-VL-7B [5] with QLoRA [6] into CrashLogic-7B, and evaluate seven decode-time variants of Temporal Contrastive Decoding (TCD). Beyond lexical metrics we score each output for *omission* and *hallucination* with an ARGUS-style dual cost [7] instantiated over our structured fields, for entailment against the forensic ground truth with an NLI cross-encoder, and for temporal grounding against prior-only baselines. All comparisons use paired tests on aligned per-video scores [8].

b) *Findings.*: Three results organise the paper.

(i) *Adaptation removes omissions, not hallucinations.* Fine-tuning reduces the omission cost from 0.462 to 0.107 (improved on 140/150 videos, $p < 10^{-4}$) and raises BLEU-4 ninefold. The hallucination cost is unchanged ($0.227 \rightarrow 0.223$, $p = 0.81$). The adapted model asserts every forensic field for every video, and roughly one field in five is wrong. On the six test videos whose ground truth states that no collision occurred, the adapted model fabricates one in three cases under greedy decoding and in all six under TCD.

(ii) *Temporal localisation is a learned prior.* CrashLogic-

7B raises mean temporal IoU from 0.012 to 0.373—but it emits exactly two crash windows, [2, 3] s or [3, 4] s, and its predicted start time is uncorrelated with the true start (Spearman $\rho=0.009$, $p=0.91$). A constant [3, 4] s window scores tIoU 0.408, higher than any model. Because frames are supplied without absolute time stamps, the model has no signal from which to localise and instead reproduces the training-set marginal—a shortcut in the sense of Geirhos *et al.* [9]. Reporting tIoU without such a baseline, as prior accident-VLM work does, overstates temporal competence.

(iii) *Temporal contrastive decoding does not repair the residue.* At $\alpha=0.5$, TCD changes no metric significantly (paired Wilcoxon $p \geq 0.21$ on all nine metrics; NLI-Score $\Delta=+0.062$, 95% CI $[-0.034, +0.156]$). At $\alpha \geq 1.5$ it significantly *increases* hallucination and omission and lowers ROUGE-L. We give a mechanistic account: a reversed-frame negative preserves every order-invariant cue—vehicle colour, weather, agent identity—so the logit difference cannot discriminate the errors that dominate the residual (wrong agent, wrong severity), and a model that has not learned to use frame order receives almost no contrast signal on timing.

c) Contributions.:

- CrashX-1500, a human-annotated dashcam crash explanation dataset: 1,500 clips with original multi-sentence forensic narratives (143,968 words) and a nine-field structured schema, released with this article.
- A held-out, leakage-checked evaluation protocol for dense crash explanation that scores omission and hallucination separately, includes prior-only temporal baselines, and reports paired statistics over per-video scores.
- CrashLogic-7B, a QLoRA adaptation of Qwen2.5-VL-7B that is the strongest system on every completeness metric, together with the finding that it leaves hallucination untouched.
- A field-level decomposition of hallucination (timestamp, severity, agent colour) with an automatic error taxonomy in the appendix.
- A paired evaluation of temporal contrastive decoding on this task, including a negative result with a mechanistic explanation and a demonstration that the collision prior worsens under contrast.
- Release of the annotation file, all model outputs, per-video scores, and regeneration scripts.

We deliberately place severity classification, structured-field accuracy, and human Likert studies outside the main results: the first two appear only as components of the hallucination analysis, and the third we could not complete at the scale a reviewer should demand (Section X). The system diagram is Figure 2.

II. RELATED WORK

a) Video-LLMs and what their benchmarks measure.:

Open Video-LLMs such as Qwen2-VL/Qwen2.5-VL [5], [10] and LLaVA-NeXT-Video [11] score well on multiple-choice suites such as Video-MME [12] and MVBench [13]. A recurring finding, however, is that these suites can be solved without

using temporal order: single-frame models match multi-frame ones on several of them [14], [15], and shuffling frames barely changes accuracy [16]. ARGUS [7] shows the complementary gap: models that verify well hallucinate heavily when asked to *generate*. Our setting is generation, our reference is a human forensic narrative, and we explicitly test for the order-insensitivity these works warn about by comparing against constant-window baselines and by measuring the correlation between predicted and true event times.

b) *Hallucination in multimodal LLMs.*: Object hallucination in captioning was formalised by CHAIR [17] and POPE [18]; HallusionBench [19] extends it to entangled language priors and visual illusions. For video, VideoHalluciner [20], EventHallusion [21], TempCompass [22], and VidHalluc [23] probe intrinsic, event-level, and temporal-sequence hallucinations, largely through paired or adversarial questions. ARGUS [7] is closest to our evaluation: it separates *hallucination* (statements not entailed by the reference) from *omission* (reference content missing from the output) for free-form captions using an LLM judge. We keep the dual-cost framing but instantiate it over the structured fields of our own CrashX-1500 annotations so that it runs without API access and yields a per-field decomposition (Section VII-A). The NLG literature has long argued that fluency and faithfulness diverge [1], [2]; Gekhman *et al.* [24] show that supervised fine-tuning on facts the model cannot ground increases hallucination in LLMs. Our omission/hallucination asymmetry (Section VI-C) is the video-explanation analogue: adaptation teaches the model to *always answer*, not to *answer correctly*.

c) *Decode-time mitigation.*: Contrastive decoding [25] sharpens a strong model’s distribution against a weaker one. VCD [4] contrasts against a visually corrupted input; ICD [26] against a perturbed instruction; M3ID [27] against the unconditioned language prior; OPERA [28] penalises over-trust on summary tokens; DoLa [29] contrasts layers. SEASON [3] brings this to video with temporally homogenised negatives and per-token self-diagnostic weights, reporting gains on VidHalluc, VideoHalluciner, and EventHallusion. Our TCD is SEASON’s temporal branch with a reversed- or shuffled-frame negative and a scalar α ; we also evaluate the full SEASON recipe. Unlike prior reports, we evaluate with paired tests on a fixed test split and find no significant benefit on this task, and we explain the null result in terms of which cues the negative preserves (Section IV-D).

d) *Temporal grounding in Video-LLMs.*: TimeChat [30], VTimeLLM [31], and LITA [32] give the language model explicit access to absolute time, via time-stamped frame tokens or dedicated time tokens, and show that this is necessary for moment localisation. Our temporal-prior finding (Section VII-B) is consistent with theirs: a model fed frames without time stamps cannot emit calibrated timestamps, and a supervised objective that demands them is satisfied by the training marginal.

e) Accident video understanding and driving VLMs.:

Dashcam accident datasets—DAD [33], A3D [34], DoTA [35], DADA-2000 [36], CCD [37]—were built for anticipation,

detection, and attention prediction, and none of them ships the dense forensic narrative that explanation evaluation requires; this is the gap our CrashX-1500 annotations fill. MM-AU [38] adds abductive cause/effect text to ego-view accidents; VRU-Accident [39] evaluates 17 MLLMs on multiple-choice QA and dense captioning of vulnerable-road-user crashes and reports that models handle visible attributes far better than causes and preventability. Driving VLMs such as DriveLM [40], DriveGPT4 [41], LingoQA [42], DriveVLM [43], and Dolphins [44] target planning and QA on nominal driving rather than post-hoc collision explanation. None of these works separates omission from hallucination, tests temporal claims against a prior-only baseline, or evaluates decode-time hallucination mitigation; CrashX does all three on a fixed split with paired statistics.

f) Faithfulness judges.: NLI cross-encoders are an established proxy for factual consistency in summarisation [45]–[47], and atomic-claim decomposition [48] is the long-form analogue. We use a DeBERTa-v3 [49] NLI cross-encoder as one of several signals and report its variance and its correlation with the structured hallucination cost, rather than treating it as ground truth.

III. TASK, DATA, AND EVALUATION PROTOCOL

A. Task

Given a dashcam clip V , a system must produce a free-text **Explanation** of the collision and, optionally, structured forensic fields. We evaluate the Explanation as the primary object and use the structured fields only to *audit* it: to decide which of the model’s claims are missing (omission) or contradicted (hallucination) relative to ground truth.

B. CrashX-1500: our human-annotated crash explanation dataset

Existing dashcam accident corpora were built for anticipation, detection, or multiple-choice question answering, and none of them provides the dense, field-aligned forensic narrative that an explanation model must be scored against. We therefore **constructed our own annotation resource**, CrashX-1500, and release it with this article as `Car_Crash_Text_Dataset_ground_truth.xlsx`.

a) Video source and annotation contribution.: The underlying visual material is 1,500 five-second dashcam clips at 10 fps drawn from the publicly available Car Crash Dataset (CCD) [37], which distributes videos with anticipation-oriented labels only. *Every textual and forensic annotation used in this article is original work produced by our annotation team*; we do not reuse any caption, explanation, severity grade, impact description, or crash-window label from prior releases. CrashX-1500 contributes 1,500 human-written multi-sentence explanations (143,968 words in total) together with a nine-field structured schema, and it is the first resource on these clips that supports separating omission from hallucination at field level.

b) Annotation schema.: Each of the 1,500 rows is annotated by a human annotator who watched the clip and filled the following fields:

- **Severity of the Crash** — minor / moderate / severe / fatal (clips with no collision receive no severity and are treated as n/a).
- **Type of Vehicles involved** — colour-and-type strings for every agent, e.g. “yellow car, black car” (647 distinct strings).
- **No. of Vehicles involved** — integer count; 0 marks a near-miss with no collision.
- **Location of impact** — free-text contact geometry per agent, e.g. “front-right corner of yellow car, front-left corner of black car” (1,335 distinct strings).
- **Start of Crash / End of Crash** — the crash window in hh:mm:ss, giving an integer-second interval $[t_s^*, t_e^*]$.
- **Explanation** — a multi-sentence causal narrative (96.0 ± 25.5 words, 5.0 sentences on average) that names the agents, the manoeuvre, the cause, and the contact point.
- **Weather Conditions** — normal / snowy / rainy / night / mist / cloudy and combinations.
- **Camera View** — the ego vehicle and, where relevant, the vehicle it follows (178 distinct descriptions).
- **Ambiguity** — an explicit note when the annotator judged some evidence uncertain (103 clips), e.g. “unclear traffic signals”, “collision not clear”.

c) Annotation protocol.: Annotators viewed each clip frame-by-frame at 10 fps and were instructed to (i) describe only what is visible, (ii) name agents by colour and type rather than by role alone, (iii) mark the crash window at the first and last frame in which contact is visible, and (iv) record any residual uncertainty in the Ambiguity field instead of guessing. The Ambiguity column is the reason we can quantify judgement calls rather than silently absorbing them into the labels. Two rows contain unparseable timestamps and are dropped, leaving 1,498 annotated clips.

d) Label distribution.: Severity is dominated by moderate (692), followed by minor (379), severe (301), and fatal (43); this 46% majority class is the constant-predictor baseline we compare against in Section VII-A. Most crashes involve two vehicles (1,228 clips), with 113 three-vehicle and 12 four- or five-vehicle events. Weather is normal in 941 clips, snowy in 410, and rainy in 122.

e) Split protocol.: We draw a stratified 80/10/10 split by severity (StratifiedShuffleSplit, seed 42): 1,198 / 150 / 150 videos (66 / 8 / 9 no-crash; Table I). Video IDs are pairwise disjoint across splits; severity proportions match to within 0.3 points. The validation split was reserved but not used for model selection in the final run (Section X).

f) No-crash clips.: Eighty-three clips are near-misses: our annotators recorded zero involved vehicles and wrote an explanation stating that no collision occurred (66 in train, 8 in validation, 9 in test). Six of the nine test explanations say so unambiguously (“fortunately, no accident occurred”); three describe an evasive manoeuvre without stating the outcome.

TABLE I

CRASHX-1500 SPLIT USED IN CRASHX (STRATIFIED 80/10/10 BY SEVERITY, SEED 42; ZERO VIDEO-ID OVERLAP). “NO-CRASH” COUNTS CLIPS OUR ANNOTATORS RECORDED WITH ZERO INVOLVED VEHICLES AND AN EXPLANATION STATING THAT NO COLLISION OCCURRED.

Split	Videos	%	No-crash	Role
Train	1,198	80.0	66	QLoRA fine-tuning
Val	150	10.0	8	Reserved (unused)
Test	150	10.0	9 (6 unambiguous)	All reported metrics
Total	1,498	100	83	2 of 1,500 rows dropped

We use the six unambiguous clips as a small but decisive probe for fabricated collisions (Section VII-C). Their ground-truth window is empty, so they contribute $\text{tIoU} = 0$ to all systems.

g) *Temporal structure of the ground truth.*: Crash windows are short and concentrated: on the 141 crash videos of the test split the window starts in $[2, 4]$ s for 80% of clips and lasts 1.55 s on average (77 windows of 1 s, 39 of 2 s, 17 of 3 s). The most frequent training window is $[3, 4]$ s (314 of 1,132 crash clips), followed by $[2, 3]$ s (253). This concentration is a property of real dashcam collisions rather than an annotation artefact, and it is exactly what makes a prior-only baseline necessary (Section III-C0d).

C. Metrics

a) *Lexical quality.*: On the Explanation field we report corpus-level BLEU-4 [50] via sacreBLEU [51], and mean per-video ROUGE-L [52], METEOR [53], CIDEr [54], and BERTScore-F1 [55] (bert-base-uncased). Because corpus BLEU and mean per-video BLEU differ (0.142 vs. 0.123 for CrashLogic-7B), we use corpus BLEU in headline tables and per-video BLEU only inside paired tests. For systems that do not emit an Explanation: field, the entire output is scored as the explanation.

b) *Entailment (NLI).*: A cross-encoder NLI model (cross-encoder/nli-deberta-v3-small [49]) scores each pair with the ground-truth forensic text as *premise* and the model output as *hypothesis*. The premise concatenates the structured fields rendered as sentences with the reference Explanation. Let P_e , P_c , P_n be entailment, contradiction, and neutral probabilities. We report

$$\text{NLI-Score} = P_e - P_c, \quad \text{NLI-Loss} = -\log P_e, \quad (1)$$

both with the Explanation as hypothesis (*NLI*) and with the *entire* output as hypothesis (*Full-NLI*). The second variant is immune to how the Explanation is extracted and therefore fairer to zero-shot systems that ignore the output format.

c) *Omission and hallucination costs.*: Following ARGUS [7], we define two normalised costs per video, but over CrashX-1500’s five structured fields $\mathcal{F} = \{\text{severity, impact, vehicles, weather, window}\}$ plus the Explanation, so that they are deterministic and reproducible without an LLM judge. Let $A \subseteq \mathcal{F}$ be the fields the model *asserts* (parsed from its output, with keyword fallbacks for unstructured text) and $G \subseteq \mathcal{F} \cup \{\text{expl}\}$ the fields the ground truth

specifies. A field $f \in A$ *conflicts* if the asserted value is disjoint from the ground truth (severity: label mismatch; impact/vehicles/weather: no shared token; window: $\text{tIoU} < 0.1$). A field $f \in G$ is *omitted* if it is not asserted or its value shares nothing with the ground truth (window: $\text{tIoU} < 0.25$; Explanation: fewer than 8 words). Then

$$C_H = \frac{|\{f \in A : f \text{ conflicts}\}|}{|A|}, \quad C_O = \frac{|\{f \in G : f \text{ omitted}\}|}{|G|}, \quad (2)$$

with $C_H=1$ if $A=\emptyset$. C_H is a *conditional* rate—the fraction of what the model chose to say that is wrong—so it is comparable across systems that assert different numbers of fields, and we always report the assertion rate $|A|/|\mathcal{F}|$ beside it. We also report per-field conflict rates (fraction of all videos with a conflict on field f), and a stricter *colour-disjoint* agent check that flags an output whose vehicle colours share nothing with the ground truth, because token overlap on generic types (“car”) makes the field-level vehicle check lenient. The colour-and-type convention in our Type of Vehicles involved field is what makes this strict check possible.

d) *Temporal grounding, with prior baselines.*: For predicted window $[\hat{t}_s, \hat{t}_e]$ and reference $[t_s^*, t_e^*]$,

$$\text{tIoU} = \frac{|[\hat{t}_s, \hat{t}_e] \cap [t_s^*, t_e^*]|}{|[\hat{t}_s, \hat{t}_e] \cup [t_s^*, t_e^*]|}, \quad (3)$$

averaged over all 150 videos (missing or unparseable windows score 0), plus hit rates $\text{THR}@0.25$ and $\text{THR}@0.50$. To interpret tIoU we add three *prior-only baselines* that ignore the video: the constant windows $[2, 3]$ s, $[3, 4]$ s (the training mode), and $[2, 4]$ s. We also report the Spearman correlation between predicted and true start times over crash videos and the multiset of distinct windows a system emits. A system whose tIoU does not exceed the constant baselines, or whose predicted starts are uncorrelated with true starts, has not localised anything.

e) *Error taxonomy.*: We code each output on crash videos with five non-exclusive failure labels plus CLEAN, using automatic proxies so that counts are reproducible: NO-NARRATIVE (under 8 words, or under 30 words with no collision verb); FLUENT-FABRICATION (≥ 30 words, no collision verb); WRONG-AGENT (colour-disjoint vehicles); WRONG-TIME (window asserted with $\text{tIoU} < 0.1$); WRONG-CAUSE (agents consistent but NLI contradiction $P_c > 0.5$); UNDER-SPECIFIC (ground truth names a colour, output names none).

The proxies are conservative for WRONG-CAUSE, which we therefore read as a lower bound.

D. Statistics

All systems are evaluated on the same 150 videos, so every comparison is paired. For each metric we report the mean, a 95% percentile bootstrap interval (2,000 resamples) [56], and for each pair of systems the mean per-video difference with a paired-bootstrap interval (5,000 resamples), a two-sided Wilcoxon signed-rank p -value [8], [57], and win/loss/tie counts. Corpus BLEU has no per-video form; its paired test uses per-video BLEU. We use $p < 0.05$ and call an effect *directional* when the paired interval includes zero.

IV. SYSTEMS

CrashX compares three families of systems under one prompt and one frame budget: zero-shot Video-LLMs, the domain-adapted CrashLogic-7B, and CrashLogic-7B decoded with Temporal Contrastive Decoding.

A. Input representation

Every system receives $K=8$ frames sampled uniformly from the 5-second clip (one every ≈ 0.7 s) and resized so that the longer side is at most 224 px, together with the prompt

“Analyze this car crash video spatiotemporally. Provide crash severity, impact point, exact timestamp window, and detailed causal explanation.”

Frames are passed to the processor as an ordered list of images. No absolute time stamps, frame indices, or frame rate are supplied—the standard usage for these models. We flag this now because it is the proximate cause of the temporal-prior effect in Section VII-B: the language model can observe frame *order* through positional encoding, but it has no signal that maps a frame to a time in seconds.

B. CrashLogic-7B: supervised domain adaptation

a) Targets.: Training targets serialise the forensic fields and the Explanation into one string:

```
Severity: ... | Impact: ... | Start:
tss | End: tes | Vehicles: ... |
NumVehicles: n | Weather: ... |
Explanation: <text>
```

The serialisation has two consequences that matter for interpretation. It teaches the model to *always* emit every field, which eliminates omission by construction and, as we show, does nothing for correctness. And it teaches the model to emit Start/End as literal seconds, an output it must produce whether or not it can perceive time.

b) Optimisation.: We fine-tune Qwen2.5-VL-7B-Instruct [5] with 4-bit NF4 quantisation and LoRA [6], [58] adapters ($r=16$, $\alpha_{\text{LoRA}}=32$, dropout 0.05) on the language model’s attention and MLP projections, minimising the token-level negative log-likelihood of the target y given frames V and prompt p :

$$\mathcal{L}(\phi) = - \sum_{t=1}^{|y|} \log p_{\theta, \phi}(y_t \mid y_{<t}, V, p), \quad (4)$$

with frozen backbone θ and trainable adapter ϕ . Training runs five epochs at learning rate 2×10^{-4} , batch size 1 with gradient accumulation 8, and a 768-token sequence cap; it takes 3.4 h on one NVIDIA A100-40GB and ends at training loss 3.09 (full hyperparameter table: Table X).

C. Temporal Contrastive Decoding

a) Decoding rule.: Let $\ell_t(V)$ be the next-token logits at step t given the original frame sequence. TCD forms a *negative view* V_{neg} of the same frames and decodes greedily from

$$\ell_t^{\text{TCD}} = (1 + \alpha) \ell_t(V) - \alpha \ell_t(V_{\text{neg}}), \quad (5)$$

which is SEASON’s contrast [3] with a fixed scalar α . $\alpha=0$ recovers greedy decoding. Our primary negative is the **reverse** of the eight frames; the **shuffle** negative applies a fixed random permutation. Both are computed with the same adapted weights, so each decoding step costs two forward passes.

b) Full SEASON.: As an ablation we run SEASON’s complete recipe: a spatially corrupted view V_S (Gaussian noise, $\sigma=0.35$), a temporally homogenised view V_T (each frame blended with the clip mean, $\beta=0.5$), and per-token self-diagnostic weights w_S, w_T from the Jensen–Shannon divergence between the frame-attention distributions of V and each negative:

$$\ell_t = (1 + \alpha) \ell_t(V) - \alpha (w_S \ell_t(V_S) + w_T \ell_t(V_T)). \quad (6)$$

D. What a reversed negative can and cannot fix

The rationale for Eq. (5) is that tokens whose probability *depends on frame order* are amplified, while tokens equally probable under either order are left alone. Write a token’s logit as a sum of an order-invariant part and an order-dependent part, $\ell_t(V) = u_t + v_t(V)$ with $v_t(V_{\text{rev}}) \approx -v_t(V)$ for cues that flip under reversal. Then Eq. (5) gives $\ell_t^{\text{TCD}} \approx u_t + (1 + 2\alpha) v_t(V)$: the contrast rescales only the order-dependent component. This yields three testable predictions, which the analysis in Section VII checks.

P1 Colour, weather, and agent identity are untouched.

These cues are visible in every frame regardless of order, so $v_t \approx 0$ for the tokens that carry them. TCD cannot reduce WRONG-AGENT errors or weather conflicts.

P2 Timing tokens are helped only if the model already uses order.

If the backbone’s window prediction is insensitive to frame order, $\ell_t(V) \approx \ell_t(V_{\text{rev}})$ on timestamp tokens and the contrast is a no-op there. A learned temporal prior therefore survives TCD.

P3 Large α amplifies noise, not signal.

Because v_t is small for most tokens, $(1 + 2\alpha)v_t$ is dominated by estimation noise at large α ; the decoded text drifts from the greedy output and both omission and hallucination should rise.

The shuffle negative destroys within-clip adjacency as well as global order and so perturbs more tokens; we expect it to move lexical metrics more than reversal without improving faithfulness.

V. EXPERIMENTAL SETUP

a) Systems.: Zero-shot systems are Qwen2.5-VL-7B and -3B [5], Qwen2-VL-2B [10], and LLaVA-NeXT-Video-7B [11]. All receive the prompt of Section IV-A; LLaVA-NeXT-Video additionally receives the sentence “Provide Severity, Impact, Start, End, Vehicles, Weather, and Explanation.” because it otherwise never emits the fields. CrashLogic-7B is decoded greedily and with TCD at $\alpha \in \{0.5, 1.0, 1.5, 2.0\}$ (reverse), $\alpha=0.5$ (shuffle), and full SEASON ($\alpha=1.0$). Prior-only baselines emit a constant window $[2, 3]$, $[3, 4]$, or $[2, 4]$ s and no text. The full system list is Table IX.

b) Decoding.: All systems decode greedily with at most 256 new tokens; TCD applies Eq. (5) at every step. Zero-shot outputs that reach the cap are scored as emitted, which penalises systems that spend tokens on preamble. Full-NLI and the omission cost are the metrics least affected by this.

c) Implementation.: PyTorch 2.6, Transformers 5.16, PEFT, and bitsandbytes 4-bit loading; inference on one A100-40GB. The NLI judge, bootstrap, and paired tests run on CPU from the released per-video outputs. A fixed seed (42) governs the split, the shuffle negative, and all resampling.

VI. RESULTS

A. Explanation quality

Table II reports lexical and embedding overlap between generated and reference explanations. Zero-shot systems cluster tightly (BLEU-4 0.016–0.026, BERTScore 0.418–0.547); the best of them, LLaVA-NeXT-Video-7B, is an order of magnitude below the adapted model on BLEU-4. Size does not explain the ordering: Qwen2.5-VL-7B trails its 3B sibling on every lexical metric because it spends its 256-token budget on markdown headers before reaching any narrative.

CrashLogic-7B reaches BLEU-4 0.142, ROUGE-L 0.336, METEOR 0.371, CIDEr 0.508, BERTScore 0.686. Paired against zero-shot Qwen2.5-VL-7B, per-video BLEU improves on 149 of 150 videos and ROUGE-L on 150 of 150 ($p < 10^{-4}$; Table V). Adding TCD at $\alpha=0.5$ leaves BLEU-4 and BERTScore unchanged to three decimals and moves ROUGE-L by +0.003 (paired 95% CI $[-0.005, +0.011]$). Fine-tuning is responsible for all lexical gains; no decoding variant changes any lexical metric significantly in the positive direction (Section VIII).

B. Entailment against the forensic ground truth

We also score each output with a DeBERTa-v3 NLI cross-encoder (full matrix in Table XI). Two readings matter. With the extracted Explanation as hypothesis, fine-tuning raises entailment mass (P_e : 0.139 $\rightarrow$ 0.224, $p=0.007$) but raises contradiction mass by more (P_c : 0.092 $\rightarrow$ 0.229, $p < 10^{-4}$), so NLI-Score alone is flat or slightly worse (Table V). Zero-shot models mostly say nothing and are scored as neutral rather than wrong. With the *entire* output as hypothesis (Full-NLI), the ordering reverses: CrashLogic-7B greedy scores 0.247 against 0.005–0.111 for zero-shot ($\Delta=+0.170$ vs. Qwen2.5-VL-7B, $p=0.0005$). We therefore treat NLI as a secondary

signal and place the primary faithfulness claims on the structured costs below.

C. Omission versus hallucination

Table III is the central result. It separates what each system fails to say from what it says wrongly, with bootstrap intervals on both.

a) Adaptation eliminates omission.: Zero-shot Qwen2.5-VL-7B omits 46% of the forensic content the ground truth specifies; CrashLogic-7B omits 11%. The paired difference is -0.356 (95% CI $[-0.382, -0.328]$, $p < 10^{-4}$), and the omission cost falls on 140 of 150 videos and rises on none. Per field (Table XII): the window is missing or badly placed in 98% of zero-shot outputs against 43% after adaptation; weather in 84% against 12%; vehicles in 19% against 2%. The serialised target format does what it was designed to do.

b) Adaptation does not reduce hallucination.: The hallucination cost of CrashLogic-7B greedy is 0.223 (95% CI $[0.200, 0.248]$). Zero-shot Qwen2.5-VL-7B scores 0.227 ($[0.188, 0.268]$). The paired difference is -0.005 ($[-0.051, +0.041]$, $p=0.81$; 67 videos better, 68 worse). The two systems assert very different amounts—the adapted model emits all five fields on every video, the zero-shot model 61% of them on average—but the fraction of asserted claims that contradict the ground truth is the same. Domain adaptation has converted silence into assertions, roughly one in five of which is false. The finding echoes Gekhman *et al.* [24] for text-only LLMs: supervised targets the model cannot ground are learned as a policy of confident answering. LLaVA-NeXT-Video-7B and the smaller Qwen models hallucinate more than either (0.285–0.564), so adaptation *is* the best available system on C_H ; the point is that it does not improve on its own backbone.

D. Temporal grounding

Table IV reports crash-window metrics with the prior-only baselines of Section III-C0d. Zero-shot systems emit a parseable window in 9–51% of videos and score tIoU ≤ 0.024 . CrashLogic-7B emits one on every video and scores 0.373 (THR@0.50 = 0.440); TCD at $\alpha=0.5$ scores 0.394 (THR@0.50 = 0.473).

The baselines reframe these numbers. The constant window $[3, 4]$ s—the most frequent training window—scores tIoU 0.408 and THR@0.50 = 0.473, matching TCD exactly on the hit rate and exceeding both adapted systems on mean tIoU. The constant window $[2, 4]$ s scores 0.518. CrashLogic-7B emits only two distinct windows, $[2, 3]$ s (88 crash videos) and $[3, 4]$ s (53), and which one it picks is unrelated to when the crash happens (Spearman $\rho=0.009$ between predicted and true start, $p=0.91$). Section VII-B examines this further. The honest summary of Table IV is that adaptation teaches the model the temporal marginal of our training split, and that tIoU without a prior baseline would have hidden this.

E. Does temporal contrastive decoding help?

Table V gives paired statistics for the key comparisons. At $\alpha=0.5$ with the reverse negative, no metric changes significantly: NLI-Score $\Delta=+0.062$ ($[-0.034, +0.156]$, $p=0.44$),

TABLE II

EXPLANATION QUALITY ON THE CRASHX-1500 TEST SPLIT ($n=150$), COMPUTED ON THE EXPLANATION FIELD (WHOLE OUTPUT FOR SYSTEMS THAT EMIT NO FIELD). BLEU-4 IS CORPUS-LEVEL; OTHER METRICS ARE PER-VIDEO MEANS. $\uparrow$: HIGHER IS BETTER. BEST PER COLUMN IN **BOLD**. NO TCD VARIANT DIFFERS SIGNIFICANTLY FROM GREEDY ON ANY COLUMN (TABLE V).

System	BLEU-4 $\uparrow$	ROUGE-L $\uparrow$	METEOR $\uparrow$	CIDEr $\uparrow$	BERTScore $\uparrow$
ZS Qwen2.5-VL-7B	0.016	0.160	0.203	0.338	0.486
ZS Qwen2.5-VL-3B	0.019	0.180	0.234	0.394	0.518
ZS Qwen2-VL-2B	0.018	0.120	0.132	0.221	0.418
ZS LLaVA-NeXT-Video-7B	0.026	0.197	0.275	0.449	0.547
CrashLogic-7B (greedy)	0.142	0.336	0.371	0.508	0.686
CrashLogic-7B + TCD ($\alpha=0.5$)	0.142	0.339	0.369	0.515	0.686
CrashLogic-7B + TCD ($\alpha=1.0$)	0.140	0.338	0.365	0.507	0.684

TABLE III

OMISSION AND HALLUCINATION COSTS (EQ. (2); $\downarrow$ BETTER) WITH 95% BOOTSTRAP INTERVALS, $n=150$. *Assert.* IS THE MEAN FRACTION OF THE FIVE FORENSIC FIELDS A SYSTEM ASSERTS; C_H IS CONDITIONAL ON ASSERTION. PAIRED TESTS VS. ZERO-SHOT QWEN2.5-VL-7B (ADAPTATION) AND VS. GREEDY (DECODING) ARE IN TABLE V.

System	Assert.	$C_O\downarrow$	$C_H\downarrow$
ZS Qwen2.5-VL-7B	0.61	0.462 [.436, .488]	0.227 [.188, .268]
ZS Qwen2.5-VL-3B	0.68	0.413 [.391, .436]	0.285 [.247, .324]
ZS Qwen2-VL-2B	0.33	0.607 [.570, .642]	0.564 [.495, .630]
ZS LLaVA-NeXT-Video-7B	0.86	0.334 [.314, .354]	0.401 [.371, .432]
CrashLogic-7B (greedy)	1.00	0.107 [.088, .127]	0.223 [.200, .248]
CrashLogic-7B + TCD $\alpha=0.5$	1.00	0.102 [.082, .122]	0.212 [.185, .236]
CrashLogic-7B + TCD $\alpha=1.0$	1.00	0.112 [.093, .132]	0.227 [.201, .255]
CrashLogic-7B + TCD $\alpha=1.5$	1.00	0.124 [.103, .148]	0.249 [.220, .276]
CrashLogic-7B + TCD $\alpha=2.0$	1.00	0.128 [.106, .150]	0.253 [.225, .282]
CrashLogic-7B + TCD shuffle	1.00	0.114 [.093, .136]	0.233 [.205, .262]
CrashLogic-7B + SEASON (full)	1.00	0.119 [.099, .140]	0.240 [.213, .267]

TABLE IV

CRASH-WINDOW GROUNDING ON THE CRASHX-1500 TEST SPLIT ($n=150$; THE 9 NO-CRASH VIDEOS SCORE $tIoU = 0$ FOR EVERY ROW). 95% BOOTSTRAP CI ON $tIoU$ IN BRACKETS. CRASHLOGIC-7B EMITS ONLY TWO WINDOWS, [2, 3] s (88 VIDEOS) AND [3, 4] s (53); PREDICTED VS. TRUE START SPEARMAN $\rho=0.009$ ($p=0.91$). ZERO-SHOT PARSE RATES ARE 0.09–0.51. CONSTANT-WINDOW BASELINES IGNORE THE VIDEO.

System	$tIoU\uparrow$	THR@.25 $\uparrow$	THR@.50 $\uparrow$
ZS Qwen2.5-VL-7B	0.012 [.000, .028]	0.000	0.000
ZS Qwen2.5-VL-3B	0.010 [.003, .018]	0.000	0.000
ZS Qwen2-VL-2B	0.000 [.000, .000]	0.000	0.000
ZS LLaVA-NeXT-Video-7B	0.024 [.007, .045]	0.000	0.000
CrashLogic-7B (greedy)	0.373 [.312, .438]	0.573	0.440
CrashLogic-7B + TCD ($\alpha=0.5$)	0.394 [.336, .456]	0.600	0.473
Constant [2, 3] s	0.366	0.567	0.433
Constant [3, 4] s (training mode)	0.408	0.593	0.473
Constant [2, 4] s	0.518	0.873	0.767

hallucination cost $\Delta=-0.011$ ($p=0.21$), omission cost $\Delta=-0.004$ ($p=0.48$), $tIoU$ $\Delta=+0.021$ ($p=0.58$). Every point estimate is in the hoped-for direction and every interval includes zero. At $\alpha=2.0$ the contrast becomes harmful: hallucination cost $+0.030$ ($p=0.031$), omission cost $+0.021$ ($p=0.026$), ROUGE-L -0.016 ($p=0.005$), Full-NLI -0.098 ($p=0.014$).

A methodological note: an unpaired normal-approximation interval with a pooled standard deviation of 0.25 would call the $\alpha=0.5$ NLI gain significant ($z \approx 2.1$). The actual per-video

standard deviation of NLI-Score is ≈ 0.55 , and the paired test does not support the claim [8]. We therefore treat mild TCD as *safe but ineffective* on this task and strong TCD as *harmful*, and we spend the next section explaining why.

VII. ANALYSIS: WHERE THE HALLUCINATIONS ARE

The aggregate costs in Table III say that adaptation does not reduce hallucination and that TCD does not either. This section asks *which* claims are wrong, tests the predictions of Section IV-D, and shows two failure modes—a learned

TABLE V

PAIRED COMPARISONS ON ALIGNED PER-VIDEO SCORES ($n=150$): MEAN DIFFERENCE (SECOND SYSTEM MINUS FIRST), PAIRED-BOOTSTRAP 95% CI (5,000 RESAMPLES), TWO-SIDED WILCOXON SIGNED-RANK p . ARROWS GIVE THE DESIRED DIRECTION. SIGNIFICANT RESULTS ($p < 0.05$) IN **BOLD**.

ADAPTATION IS SIGNIFICANT ON EVERY COMPLETENESS METRIC, NON-SIGNIFICANT ON C_H , AND SIGNIFICANTLY *worse* ON NLI CONTRADICTION MASS. EVERY TCD RESULT AT $\alpha=0.5$ IS NON-SIGNIFICANT; EVERY SIGNIFICANT TCD RESULT AT $\alpha=2.0$ IS A DEGRADATION. REMAINING VARIANTS ARE IN TABLE III AND TABLE VII.

Metric	ZS Qwen2.5-VL-7B $\rightarrow$ CrashLogic-7B greedy			Greedy $\rightarrow$ TCD $\alpha=0.5$ (reverse)			Greedy $\rightarrow$ TCD $\alpha=2.0$ (reverse)		
	Δ	95% CI	p	Δ	95% CI	p	Δ	95% CI	p
BLEU-4 (per-video) $\uparrow$	+0.108	[+.098, +.117]	$< 10^{-4}$	+0.000	[−.009, +.009]	0.44	−0.009	[−.019, +.000]	0.14
ROUGE-L $\uparrow$	+0.176	[+.163, +.189]	$< 10^{-4}$	+0.003	[−.005, +.011]	0.22	−0.016	[−.026, −.006]	0.005
fIoU $\uparrow$	+0.360	[+.297, +.426]	$< 10^{-4}$	+0.021	[−.044, +.084]	0.58	−0.017	[−.083, +.046]	0.58
C_O (omission) $\downarrow$	−0.356	[−.382, −.328]	$< 10^{-4}$	−0.004	[−.017, +.008]	0.48	+0.021	[+.002, +.040]	0.026
C_H (hallucination) $\downarrow$	−0.005	[−.051, +.041]	0.81	−0.011	[−.027, +.005]	0.21	+0.030	[+.005, +.057]	0.031
NLI P_e $\uparrow$	+0.084	[+.019, +.148]	0.007	+0.026	[−.029, +.080]	0.24	−0.005	[−.063, +.055]	0.51
NLI P_c $\downarrow$	+0.137	[+.069, +.203]	$< 10^{-4}$	−0.036	[−.094, +.022]	0.94	−0.027	[−.089, +.034]	0.58
NLI-Score $\uparrow$	−0.052	[−.162, +.055]	0.41	+0.062	[−.034, +.156]	0.44	+0.021	[−.075, +.119]	0.91
Full-NLI-Score $\uparrow$	+0.170	[+.077, +.269]	0.0005	−0.012	[−.104, +.081]	0.77	−0.098	[−.186, −.013]	0.014

temporal prior and a collision prior—that aggregate metrics conceal.

A. Field-level decomposition

Table VI summarises the conflict rates that dominate C_H ; the full matrix is Table XII.

a) *Where adapted models are wrong.*: For CrashLogic-7B greedy, two fields carry almost all of C_H : the crash window conflicts on 42.7% of videos and severity on 54.0%. Weather conflicts on 12.0%; impact side and vehicle type almost never ($\leq 2\%$). Under the strict colour-disjoint check—does the output name any vehicle colour that appears in the ground truth?—21.3% of crash videos receive a contradicting colour. Adaptation makes the model name colours far more often and only slightly more accurately, which is why the aggregate C_H is flat.

b) *P1 holds: TCD does not touch order-invariant fields.*: Colour-disjoint rates stay within [20.6%, 29.8%] across decoding variants; weather within [11.3%, 13.3%]. Only $\alpha \geq 1.5$ moves these rates, and it moves them *up*. A reversed clip shows the same black car in the same rain; contrasting against it cannot make the model see a different colour.

c) *Severity is a labelling problem, not a perception problem.*: The 54% severity conflict rate means CrashLogic-7B is correct on 46% of videos—no better than a constant moderate predictor (46.0% of the test split). We include severity in C_H because the model asserts it as fact, but we do not read this rate as evidence of visual hallucination.

B. The temporal prior

Figure 1 plots the distribution of ground-truth crash starts against the starts CrashLogic-7B predicts. The ground truth spans [0, 5] s with a mode at 2 s; the model’s predictions occupy exactly two values.

Three facts establish that this is a prior rather than perception. First, the two emitted windows are the two most frequent training windows ([3, 4] s: 314 clips; [2, 3] s: 253 clips), and every emitted window is exactly one second long.

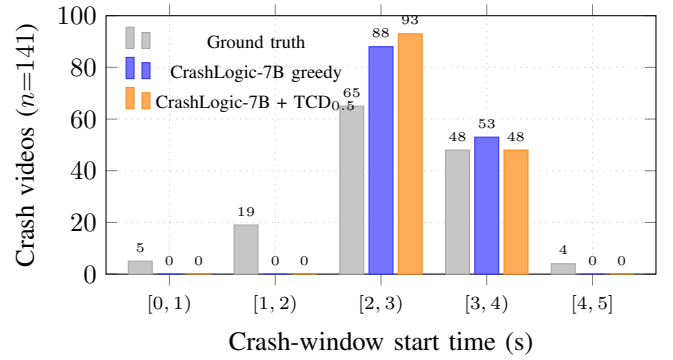


Fig. 1. The temporal prior. Ground-truth crash starts span the whole clip; the adapted model predicts only 2 s or 3 s, always with a 1-second window, matching the two most frequent training windows. Predicted and true starts are uncorrelated (Spearman $\rho=0.009$ greedy, 0.028 TCD), and a constant [3, 4] s window scores higher fIoU than either system (Table IV).

Second, the predicted start is uncorrelated with the true start ($\rho=0.009$). Third, the constant window [3, 4] s beats the model (Table IV). The model has learned *when crashes usually happen in CrashX-1500*, not *when this crash happened*.

Frames are supplied without absolute time (Section IV-A), so under Eq. (4) the loss on Start:/End: tokens is minimised by the marginal. Time-aware Video-LLMs that inject time stamps or time tokens [30]–[32] exist precisely because this shortcut is otherwise unavoidable.

a) *P2 holds: TCD cannot fix a prior.*: TCD at $\alpha=0.5$ changes the emitted window on 29% of videos—but only between [2, 3] s and [3, 4] s—fixing 16 window conflicts and introducing 12 (Table XV). No variant creates a third window or induces correlation with the true start ($\rho \leq 0.03$).

C. The collision prior: fabricated crashes on no-crash videos

Six test videos are near-misses whose ground truth states explicitly that no collision occurred. CrashLogic-7B greedy fabricates a collision on 3 of 6 and hedges on the other 3; under

TABLE VI

WHERE HALLUCINATIONS CONCENTRATE ($n=150$). *Conflict* RATE (%): FIELD ASSERTED AND WRONG (WINDOW: tIoU < 0.1). *Colour-disjoint*: OUTPUT NAMES A VEHICLE COLOUR THAT NEVER APPEARS IN THE GROUND TRUTH (141 CRASH VIDEOS; PARENTHESES: RATE AMONG VIDEOS THAT NAME ANY COLOUR). WINDOW AND SEVERITY DOMINATE; COLOUR ERRORS STAY FLAT UNDER MILD TCD (P1). FULL FIELD-WISE RATES ARE IN TABLE XII.

System	Window	Severity	Colour-disjoint	Weather
ZS Qwen2.5-VL-7B	26.0	9.3	14.2 (33)	14.0
ZS LLaVA-NeXT-Video-7B	45.3	57.3	7.1 (20)	58.7
CrashLogic-7B (greedy)	42.7	54.0	21.3 (22)	12.0
CrashLogic-7B + TCD $\alpha=0.5$	40.0	51.3	20.6 (21)	12.7
CrashLogic-7B + TCD $\alpha=2.0$	44.0	60.0	29.8 (31)	12.0

TCD at $\alpha=0.5$ and 1.0 it fabricates on **6 of 6** (Table XIII). No system ever predicts severity n/a. Reversing a near-miss clip produces a sequence in which vehicles converge rather than diverge, so the contrast *amplifies* collision-like continuations. The prompt itself (“analyze this car crash video”) presupposes a crash; we return to this in Section X.

D. Error profile and what TCD changes

The automatic taxonomy (Table XIV) shows the same asymmetry in codes: adaptation removes NO-NARRATIVE (10.6% $\rightarrow$ 0%) and UNDER-SPECIFIC (40.4% $\rightarrow$ 0%), while roughly doubling WRONG-AGENT (9.9% $\rightarrow$ 21.3%) and WRONG-TIME (19.9% $\rightarrow$ 39.0%). TCD at $\alpha=0.5$ rewrites the free-text Explanation on every video (ROUGE-L to greedy 0.48) while changing the vehicle set on only 39% of videos and the severity on 10% (Table XV): the structured fields where hallucination lives are largely inherited from greedy decoding. As α grows, both costs rise (**P3**). Per-video NLI-Score has standard deviation 0.55 and correlates weakly with $\mathcal{C}_H$ (Spearman -0.10), so we keep primary faithfulness claims on the structured costs.

VIII. DECODING ABLATION

We fix the CrashLogic-7B checkpoint and vary only the decoding rule. Table VII summarises the three configurations that matter for the claim; the full matrix (all α , shuffle, SEASON) is Table XVII.

a) Contrast strength.: From greedy to $\alpha=0.5$ every faithfulness metric moves in the desired direction by a small, non-significant amount. At $\alpha=2.0$ hallucination, omission, ROUGE-L, and Full-NLI all degrade significantly (Table V), and the colour-disjoint agent error rises from 21% to 30% (Table VI). The trajectory matches prediction P3: beyond $\alpha \approx 1$ the amplified component is mostly noise.

b) Shuffle and full SEASON.: Neither alternative beats reverse TCD on any faithfulness metric (Table XVII). Shuffle posts the best lexical numbers and the worst NLI-Score; full SEASON raises both costs relative to greedy. The failure to move $\mathcal{C}_H$ is therefore not a tuning problem: seven configurations spanning two negative constructions leave colour, weather, and severity conflict rates where greedy decoding put them.

c) What the ablation establishes.: Mild temporal contrast is safe but ineffective; strong contrast is harmful; no recipe tested here reduces hallucination on this task.

IX. QUALITATIVE ANALYSIS

Table VIII shows three systems on two test videos that illustrate the omission/hallucination asymmetry; a no-crash example is in Table XVI.

a) Video 001227—adaptation recovers the narrative.

The reference describes an oncoming white car on a curved two-lane road and a front-left to front-left impact after an evasive turn. Zero-shot Qwen2.5-VL-7B emits markdown scaffolding and no narrative: NO-NARRATIVE, ROUGE-L 0. CrashLogic-7B names the white oncoming car, a loss of control, and a front/left impact; agents and cause family match (ROUGE-L 0.45). TCD keeps the agents and edges closer to the reference wording (0.50). Both adapted outputs place the crash at [2, 3] s; the reference window is [3, 4] s (tIoU 0).

b) Video 000104—adaptation hallucinates the agent.

The reference: the camera car strikes the rear of a slow *black truck*; a red bus passes nearby. Zero-shot text discusses weather and never reaches a collision. CrashLogic-7B constructs a coherent rear-end story—but with a *red car* as the lead vehicle: WRONG-AGENT. TCD keeps the red car and adds unsupported parked cars and pedestrians. Under BLEU/ROUGE the adapted outputs score 0.41 and 0.31 against 0.09 for zero-shot, so lexical metrics rank a specific wrong explanation far above a vague one. This is the case that motivates scoring hallucination separately from omission.

X. LIMITATIONS AND THREATS TO VALIDITY

a) Test-set size.: One hundred and fifty videos is enough to detect the adaptation effects ($p < 10^{-4}$ on every completeness metric) and to rule out large TCD effects, but not to detect small ones: the paired interval on NLI-Score at $\alpha=0.5$ spans $[-0.034, +0.156]$. A TCD gain of $+0.05$ NLI-Score or -0.01 $\mathcal{C}_H$ would require roughly four times the data to establish. The no-crash probe uses six videos; we report counts, not rates, and draw no inference beyond “fabrication occurs and worsens under contrast”.

b) Annotation quality of CrashX-1500.: Each clip was annotated once, so we do not report inter-annotator agreement, and the Explanation field is a single annotator’s reading of the event rather than a consensus narrative. Free-text fields

TABLE VII

DECODING ABLATION ON THE FIXED CRASHLOGIC-7B CHECKPOINT ($n=150$). MILD REVERSE TCD IS SAFE BUT NEVER SIGNIFICANTLY BETTER THAN GREEDY; STRONG CONTRAST SIGNIFICANTLY HURTS (TABLE V). SHUFFLE AND FULL SEASON ROWS ARE IN TABLE XVII. [†] MARKS A SIGNIFICANT DEGRADATION VS. GREEDY ($p < 0.05$).

Variant	ROUGE-L [†]	Full-NLI [†]	C_H ↓	C_O ↓
Greedy ($\alpha=0$)	0.336	0.247	0.223	0.107
TCD $\alpha=0.5$ (reverse)	0.339	0.234	0.212	0.102
TCD $\alpha=2.0$ (reverse)	0.319 [†]	0.148 [†]	0.253 [†]	0.128 [†]

TABLE VIII

QUALITATIVE COMPARISON ON TWO HELD-OUT TEST VIDEOS. EXPLANATIONS ARE LIGHTLY TRUNCATED; STRUCTURED FIELDS ARE THE SYSTEMS' OWN. PER-VIDEO ROUGE-L (R-L) AGAINST THE REFERENCE EXPLANATION IS GIVEN FOR EACH OUTPUT. **Red** MARKS A CLAIM THAT CONTRADICTS THE REFERENCE. A NO-CRASH EXAMPLE (COLLISION PRIOR) IS DISCUSSED IN SECTION VII-C AND TABLE XVI.

	Video 001227 — <i>adaptation recovers the narrative</i>	Video 000104 — <i>adaptation hallucinates the agent</i>
Reference	Curved two-lane road; a <i>white car</i> approaches from the opposite direction; the camera car turns left to avoid it; front-left to front-left collision. Severity moderate; window [3, 4] s.	Busy commercial road in daylight; a red bus passes; a slow <i>black truck</i> ahead; the front of the camera car strikes the rear of the black truck. Severity moderate; window [1, 5] s.
Zero-shot Qwen2.5-VL-7B	“### Analysis of the Car Crash Video — Video Details: Date: 20/06/2014 — Time: 17:25:23 — Camera Perspective: ...” No narrative reaches the crash. NO-NARRATIVE. R-L 0.00.	“...overcast weather conditions ... The road appears to be clear with no visible obstacles or hazards. Traffic seems” (cut at 256 tokens). No collision described. FLUENT-FABRICATION. R-L 0.09.
CrashLogic-7B (greedy)	Severity severe Start 2.0 End 3.0 Vehicles: White Car, Camera Car. “...a white car appeared from the opposite direction, speeding recklessly ... lost control ... impact between the front of the white car and the left side of the camera car.” Agents and cause family match. R-L 0.45, tIoU 0.	Severity moderate Start 2.0 End 3.0 Vehicles: Camera Car, Red Car . “...the camera car followed closely behind a red car ... the red car abruptly applied its brakes ... the front of the camera car collided with the rear of the red car.” Coherent rear-end story, wrong agent. WRONG-AGENT. R-L 0.41, tIoU 0.25.
CrashLogic-7B + TCD_{0.5}	Severity severe Start 2.0 End 3.0 Vehicles: Camera Car, White Car. “...a white car approached from the opposite direction on the left side ... both vehicles attempted to cross the intersection, resulting in a collision between the front-left of the camera car and the front-right of the white car.” R-L 0.50, tIoU 0.	Severity moderate Start 2.0 End 3.0 Vehicles: Camera Car, Red Car . “...several cars were parked, while a few pedestrians were walking ... As the camera car attempted to overtake the black car in front of it, it encountered a red car that had come to a stop ...” Same wrong agent; new unsupported details. R-L 0.31, tIoU 0.25.

such as Location of impact (1,335 distinct strings) and Camera View (178) are expressive but not drawn from a closed vocabulary, which is why our automatic checks fall back on token overlap. The Ambiguity column records the 103 clips on which the annotator flagged uncertain evidence; a double-annotated subset with adjudication is the natural next release.

c) Automatic judges.: The structured costs depend on regular-expression field extraction and token-overlap matching, which we chose for determinism. They are lenient on vehicle *type* (any shared token counts) and strict on colour; we report both. The NLI cross-encoder abstains on most pairs and correlates weakly with the structured cost (Spearman -0.10 with C_H). An LLM judge in the ARGUS style [7] would give sentence-level hallucination and omission rates on the free text itself; we did not use one because we wanted every number reproducible offline, and because judge-model choice would add a second source of variance to a study whose main point is variance.

d) No human evaluation.: We prepared a blinded 200-item, three-annotator Likert protocol (temporal accuracy, faithfulness, explanation quality) and release the item pack, but the annotations were not completed at the scale the protocol requires. The structured costs are therefore the paper’s faithfulness ground truth. Human ratings would most likely sharpen, not reverse, the omission/hallucination asymmetry, since the

fabricated collisions in Section VII-C are unambiguous to a human viewer.

e) Frame budget and time information.: Eight frames at ≤ 224 px and no absolute time stamps is the standard, cheap usage of these models and the setting our compute allowed, but it is also the reason the temporal prior arises. We have shown that the prior exists and beats the model; we have *not* shown that a time-aware input [30]–[32], denser sampling, or a time-token objective would remove it on CrashX-1500. That is the obvious next experiment.

f) Prompt presupposition.: The prompt asks the model to “analyze this car crash video”. For the 5.5% of clips that are near-misses this presupposes a false fact. We kept the prompt identical for all systems because changing it would confound the comparison, but the fabrication counts in Table XIII are an upper bound on what a neutrally phrased prompt would produce.

g) Validation split and model selection.: The 150-video validation split was reserved but not used for checkpoint selection; the final checkpoint is the last of five epochs. No hyperparameter of CrashLogic-7B or TCD was tuned on the test split— $\alpha=0.5$ was chosen before the paired tests were run, and the paired tests then declined to support it.

h) Single backbone, single seed.: All adapted systems derive from one QLoRA run of one 7B backbone. Variance across training seeds is unmeasured, and stronger or time-aware backbones may not share the temporal prior. The zero-

shot comparison spans four models from two families and the hallucination cost is high for all of them, which suggests the omission/hallucination asymmetry is not specific to Qwen2.5-VL, but we have adapted only one.

i) *No proprietary systems.*: GPT-4o, Gemini, and similar API models are absent. VRU-Accident [39] reports that Gemini-1.5-Flash leads its dense-captioning leaderboard, and ARGUS [7] reports that Gemini-2.0-Flash still hallucinates in about 41% of caption sentences; we expect proprietary systems to reduce omission substantially and hallucination less, but that is a prediction, not a result.

j) *Simplified SEASON.*: Our temporal homogenisation acts in pixel space rather than on vision-encoder features, and our frame-attention estimate is a uniform partition of the key sequence. The full-SEASON row is therefore a faithful *implementation of the recipe* on this backbone, not a reproduction of the original system.

XI. CONCLUSION

We set out to measure whether Video-LLMs can explain dashcam collisions faithfully, and we separated the two ways they can fail. Zero-shot models omit: on CrashX-1500 they leave out nearly half of the forensic content and often never mention the crash. Domain adaptation with QLoRA removes the omissions almost entirely and lifts every lexical metric by a wide, statistically unambiguous margin. It does not touch hallucination. The adapted model asserts every field on every video, one in five of those assertions contradicts the ground truth, one in five explanations names a vehicle of the wrong colour, and every confirmed no-crash video receives a fabricated collision under the decoding variants that score best on lexical metrics.

We also showed that the adapted model’s crash-window predictions are a reproduction of the training-set marginal—two windows, no correlation with the true onset, beaten by a constant—and that this arises because the model is never told what time a frame is. Temporal IoU reported without a prior-only baseline would have concealed this, and we suggest that such baselines become standard in accident-VLM evaluation.

Finally, we tested the natural training-free remedy and found that it does not work here, and we explained why: a reversed-frame negative preserves every order-invariant cue, so it cannot reach the errors that dominate—wrong colours, wrong labels, and a prior that does not depend on order to begin with. Mild contrast is harmless and nudges causal phrasing in the right direction; strong contrast makes both omission and hallucination significantly worse.

The constructive message is about what to fix next. Omission is a data and format problem, and it is solved. Hallucination on this task has three identifiable sources—no time signal, a prompt that presupposes a crash, and a supervised objective that rewards confident assertion—and each suggests a specific intervention: time-stamped inputs or time tokens, a no-crash-aware prompt and target, and a training signal that penalises unsupported claims rather than rewarding their presence. We

release all outputs, per-video scores, and evaluation code so that those interventions can be measured against the same bar.

APPENDIX

This appendix collects setup tables, the full NLI and field-wise matrices, the no-crash probe, the automatic error taxonomy, the complete decoding ablation, and the system diagram. None of these change the three main findings in the body.

A. *Systems and hyperparameters*

B. *Full NLI and field matrices*

C. *No-crash probe, taxonomy, and TCD divergence*

D. *Full decoding ablation*

REPRODUCIBILITY STATEMENT

The CrashX-1500 annotation file, code, data splits, all 1,650 model outputs (11 systems $\times$ 150 test videos), per-video metric vectors, and the scripts that regenerate every table and figure in this paper are released at <https://github.com/harshalDharpure/Crash-X>. No number in this paper requires API access or GPU inference to reproduce from the released outputs.

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TABLE IX

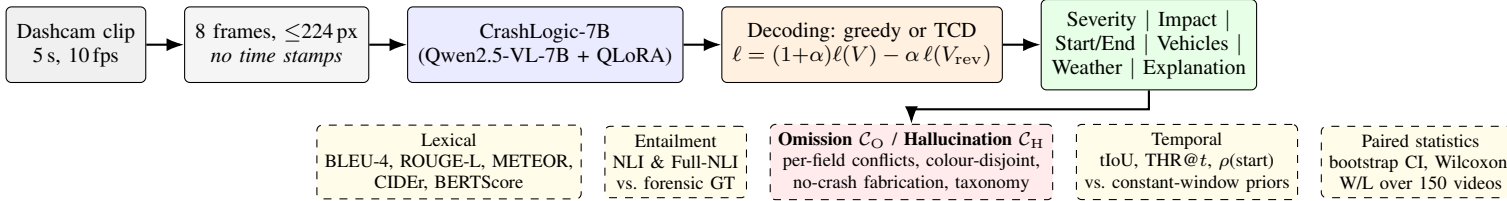
SYSTEMS EVALUATED ON THE CRASHX-1500 TEST SPLIT ($n=150$). ALL VIDEO-LLMS RECEIVE THE SAME PROMPT AND THE SAME 8 UNIFORMLY SAMPLED FRAMES (≤ 224 px). PRIOR-ONLY BASELINES EMIT A CONSTANT CRASH WINDOW AND NO TEXT.

System	Size	Adaptation / decoding
Qwen2.5-VL [5]	7B	Zero-shot, greedy
Qwen2.5-VL [5]	3B	Zero-shot, greedy
Qwen2-VL [10]	2B	Zero-shot, greedy
LLaVA-NeXT-Video [11]	7B	Zero-shot, greedy
CrashLogic-7B (ours)	7B+LoRA	QLoRA on 1,198 CrashX-1500 clips; greedy
+TCD reverse	7B+LoRA	Eq. (5), $\alpha \in \{0.5, 1, 1.5, 2\}$
+TCD shuffle	7B+LoRA	Eq. (5), $\alpha=0.5$
+SEASON (full)	7B+LoRA	Eq. (6), $\alpha=1$
Constant window	prior	$[2, 3]$ s, $[3, 4]$ s, or $[2, 4]$ s

TABLE X

CRASHLOGIC-7B TRAINING AND INFERENCE SETTINGS.

Setting	Value
Backbone	Qwen2.5-VL-7B-Instruct
Quantisation	4-bit NF4 (bitsandbytes)
Adapter	LoRA $r=16$, $\alpha_{\text{LoRA}}=32$, dropout 0.05
Learning rate / schedule	2×10^{-4} , cosine
Epochs / batch	5 / 1 (grad. accum. 8)
Keyframes / max side	8 / 224 px, no time stamps
Max sequence length	768 tokens
Hardware / wall-clock	$1 \times \text{A100-40GB}$ / 3.4 h
Final training loss	3.09
Decoding	Greedy; max 256 new tokens
TCD	Eq. (5); reverse or shuffle negative



Zero-shot comparators (same frames, same prompt): Qwen2.5-VL-7B/3B, Qwen2-VL-2B, LLaVA-NeXT-Video-7B.

Fig. 2. CrashX systems and evaluation stack. CrashLogic-7B is a QLoRA adaptation of Qwen2.5-VL-7B trained to emit serialised forensic fields and a free-text Explanation; Temporal Contrastive Decoding (TCD) optionally contrasts its logits against a reversed-frame negative at decode time. Every output is scored for lexical overlap, entailment, separate omission and hallucination costs with a field-level decomposition, and temporal grounding against prior-only baselines, with all comparisons paired over the same 150 test videos.

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TABLE XI

ENTAILMENT OF MODEL OUTPUTS BY THE FORENSIC GROUND TRUTH ($n=150$; CROSS-ENCODER NLI [49], PREMISE = STRUCTURED FIELDS + REFERENCE EXPLANATION). *NLI* USES THE EXTRACTED EXPLANATION AS HYPOTHESIS; *Full-NLI* USES THE ENTIRE OUTPUT AND IS INDEPENDENT OF FIELD EXTRACTION. $NLI\text{-SCORE} = P_e - P_c$ ($\uparrow$); $NLI\text{-LOSS} = -\log P_e$ ($\downarrow$). 95% BOOTSTRAP INTERVALS IN BRACKETS. BEST PER COLUMN IN **BOLD**.

System	NLI (Explanation as hypothesis)					Full-NLI (whole output)	
	$P_e \uparrow$	$P_c \downarrow$	P_n	Score $\uparrow$	Loss $\downarrow$	$P_e \uparrow$	Score $\uparrow$
ZS Qwen2.5-VL-7B	0.139	0.092	0.769	0.047 [-.019, .110]	3.811	0.128	0.076 [.028, .130]
ZS Qwen2.5-VL-3B	0.145	0.056	0.799	0.089 [.040, .138]	3.562	0.152	0.111 [.064, .159]
ZS Qwen2-VL-2B	0.092	0.099	0.809	-0.007 [-.066, .050]	4.462	0.099	0.005 [-.055, .062]
ZS LLaVA-NeXT-Video-7B	0.157	0.061	0.782	0.097 [.041, .155]	3.464	0.153	0.092 [.035, .151]
CrashLogic-7B (greedy)	0.224	0.229	0.548	-0.005 [-.092, .090]	3.124	0.402	0.247 [.156, .333]
CrashLogic-7B + TCD ($\alpha=0.5$)	0.249	0.193	0.558	0.057 [-.031, .146]	2.944	0.395	0.234 [.134, .327]
CrashLogic-7B + TCD ($\alpha=1.0$)	0.276	0.186	0.538	0.090 [.003, .180]	2.954	0.386	0.237 [.142, .325]

TABLE XII

FIELD-LEVEL CONFLICT AND OMISSION RATES FOR ALL SYSTEMS (SUPPLEMENT TO TABLE VI).

System	Conflict rate (%)						Omission rate (%)					
	Window	Severity	Weather	Impact	Vehicle	Colour-disjoint (%)	Window	Severity	Weather	Impact	Vehicle	Expl.
ZS Qwen2.5-VL-7B	26.0	9.3	14.0	20.0	5.3	14.2 (33)	98.0	24.7	84.0	32.7	19.3	7.3
ZS Qwen2.5-VL-3B	37.3	28.7	13.3	19.3	4.0	18.4 (40)	99.3	22.7	81.3	24.0	8.7	0.7
ZS Qwen2-VL-2B	8.7	21.3	6.7	6.0	2.7	0.7 (17)	100	60.0	86.0	62.7	42.7	0.7
ZS LLaVA-NeXT-Video-7B	45.3	57.3	58.7	11.3	2.7	7.1 (20)	96.7	0.7	70.7	19.3	2.7	0.7
CrashLogic-7B (greedy)	42.7	54.0	12.0	0.7	2.0	21.3 (22)	42.7	0.0	12.0	0.7	2.0	0.0
TCD $\alpha=0.5$ (reverse)	40.0	51.3	12.7	0.7	1.3	20.6 (21)	40.0	0.0	12.7	0.7	1.3	0.0
TCD $\alpha=1.0$ (reverse)	42.7	53.3	11.3	2.0	4.0	20.6 (21)	42.7	0.0	11.3	2.0	4.7	0.0
TCD $\alpha=1.5$ (reverse)	45.3	57.3	12.7	2.7	4.0	27.0 (28)	45.3	0.0	12.7	4.0	6.0	0.0
TCD $\alpha=2.0$ (reverse)	44.0	60.0	12.0	2.7	4.0	29.8 (31)	44.7	0.0	12.0	6.0	7.3	0.0
TCD $\alpha=0.5$ (shuffle)	40.7	55.3	13.3	2.7	2.7	23.4 (24)	40.7	0.0	13.3	4.0	4.7	0.0
SEASON (full)	45.3	55.3	12.7	2.7	3.3	23.4 (24)	45.3	0.0	12.7	3.3	3.3	0.0

TABLE XIII

FABRICATED COLLISIONS ON THE SIX TEST VIDEOS WHOSE REFERENCE STATES THAT NO COLLISION OCCURRED. *Fabricated*: THE EXPLANATION NARRATES A COLLISION AND CONTAINS NO HEDGE (“AVOIDED”, “NO ACCIDENT”). *Hedged*: THE EXPLANATION CONTAINS A HEDGE. COUNTS OUT OF 6. NO SYSTEM PREDICTS SEVERITY N/A.

System	Fabricated	Hedged
ZS Qwen2.5-VL-7B	3	1
ZS Qwen2.5-VL-3B	5	0
ZS Qwen2-VL-2B	4	0
ZS LLaVA-NeXT-Video-7B	5	1
CrashLogic-7B (greedy)	3	3
TCD $\alpha=0.5$ (reverse)	6	0
TCD $\alpha=1.0$ (reverse)	6	0
TCD $\alpha=1.5$ (reverse)	5	1
TCD $\alpha=2.0$ (reverse)	1	5
TCD $\alpha=0.5$ (shuffle)	5	1
SEASON (full)	5	1

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TABLE XIV

AUTOMATIC ERROR TAXONOMY ON THE 141 CRASH VIDEOS (PERCENT OF VIDEOS; CODES ARE NOT MUTUALLY EXCLUSIVE, SO ROWS NEED NOT SUM TO 100). CLEAN = NO CODE FIRED. ADAPTATION ELIMINATES THE TWO *omission-type* CODES AND ROUGHLY DOUBLES THE TWO *hallucination-type* CODES; TCD AT $\alpha=0.5$ LOWERS ONLY WRONG-CAUSE.

System	NO-NARR.	UNDER-SP.	WRONG-AG.	WRONG-TIME	WRONG-CAUSE	FLUENT-FAB.	CLEAN $\uparrow$
ZS Qwen2.5-VL-7B	10.6	40.4	9.9	19.9	5.7	9.9	23.4
ZS Qwen2.5-VL-3B	4.3	51.1	13.5	32.6	2.1	3.5	19.1
ZS Qwen2-VL-2B	42.6	52.5	0.0	9.2	5.7	0.0	3.5
ZS LLaVA-NeXT-Video-7B	0.7	65.2	5.7	44.7	5.7	0.0	12.8
CrashLogic-7B (greedy)	0.0	0.0	21.3	39.0	17.7	1.4	38.3
TCD $\alpha=0.5$ (reverse)	0.0	0.0	20.6	36.2	10.6	0.7	45.4
TCD $\alpha=1.0$ (reverse)	0.0	1.4	20.6	39.0	11.3	1.4	42.6
TCD $\alpha=1.5$ (reverse)	0.0	1.4	26.2	40.4	12.1	3.5	37.6
TCD $\alpha=2.0$ (reverse)	0.0	2.8	27.7	36.2	9.9	5.7	36.2
TCD $\alpha=0.5$ (shuffle)	0.0	2.1	22.7	36.2	17.0	0.7	41.1
SEASON (full)	0.0	0.0	23.4	41.1	14.2	0.7	38.3

TABLE XV

HOW FAR EACH DECODING VARIANT MOVES FROM THE GREEDY OUTPUT ON THE SAME VIDEO ($n=150$). *Window / Severity / Vehicles changed*: FRACTION OF VIDEOS ON WHICH THE PARSED FIELD DIFFERS FROM GREEDY. *R-L to greedy*: ROUGE-L BETWEEN THE VARIANT’S AND GREEDY’S EXPLANATION (THE FREE TEXT IS REWRITTEN ON 100% OF VIDEOS FOR EVERY VARIANT). *Window fixed / broken*: WINDOW CONFLICTS ($\text{tIoU} < 0.1$) REMOVED / INTRODUCED RELATIVE TO GREEDY’S 64.

Variant	Window changed	Severity changed	Vehicles changed	R-L to greedy	Window fixed / broken
TCD $\alpha=0.5$ (reverse)	0.29	0.10	0.39	0.48	16 / 12
TCD $\alpha=1.0$ (reverse)	0.35	0.19	0.50	0.47	17 / 17
TCD $\alpha=1.5$ (reverse)	0.39	0.28	0.59	0.43	15 / 19
TCD $\alpha=2.0$ (reverse)	0.41	0.32	0.62	0.42	15 / 17
TCD $\alpha=0.5$ (shuffle)	0.29	0.23	0.54	0.46	13 / 10
SEASON (full)	0.27	0.17	0.53	0.47	10 / 14

TABLE XVI

QUALITATIVE COMPARISON ON THREE HELD-OUT TEST VIDEOS, ONE PER FINDING. EXPLANATIONS ARE LIGHTLY TRUNCATED; STRUCTURED FIELDS ARE THE SYSTEMS’ OWN. PER-VIDEO ROUGE-L (R-L) AGAINST THE REFERENCE EXPLANATION IS GIVEN FOR EACH OUTPUT. **Red** MARKS A CLAIM THAT CONTRADICTS THE REFERENCE.

	Video 001227 — adaptation recovers the narrative	Video 000104 — adaptation hallucinates the agent	Video 000661 — the collision prior
Reference	Curved two-lane road; a <i>white car</i> approaches from the opposite direction; the camera car turns left to avoid it; front-left to front-left collision. Severity moderate; window [3, 4]s.	Busy commercial road in daylight; a red bus passes; a slow <i>black truck</i> ahead; the front of the camera car strikes the rear of the black truck. Severity moderate; window [1, 5]s.	Busy highway; a white truck on the far right; a red car attempts to overtake it, “but fortunately, <i>no accidents took place</i> .” Severity n/a; no window.
Zero-shot Qwen2.5-VL-7B	“### Analysis of the Car Crash Video — Video Details: Date: 20/06/2014 — Time: 17:25:23 — Camera Perspective: ...” No narrative reaches the crash. NO-NARRATIVE. R-L 0.00.	“...overcast weather conditions ... The road appears to be clear with no visible obstacles or hazards. Traffic seems” (cut at 256 tokens). No collision described. FLUENT-FABRICATION. R-L 0.09.	“...At approximately 0:05 seconds there is a sudden jolt forward, indicating a collision ... the red sedan ... skidding ... moderate to severe .” Fabricated crash. R-L 0.12.
CrashLogic-7B (greedy)	Severity severe Start 2.0 End 3.0 Vehicles: White Car, Camera Car. “...a white car appeared from the opposite direction, speeding recklessly ... lost control ... impact between the front of the white car and the left side of the camera car.” Agents and cause family match. R-L 0.45, tIoU 0.	Severity moderate Start 2.0 End 3.0 Vehicles: Camera Car, Red Car . “...the camera car followed closely behind a red car ... the red car abruptly applied its brakes ... the front of the camera car collided with the rear of the red car.” Coherent rear-end story, wrong agent. WRONG-AGENT. R-L 0.41, tIoU 0.25.	Severity moderate Start 2.0 End 3.0 Vehicles: White Truck, Red Car. “...the red car made a left turn without noticing the white truck. As a result, the white truck collided with the rear left side of the red car ...” Fabricated collision with correct agents. R-L 0.28.
CrashLogic-7B + TCD_{0.5}	Severity severe Start 2.0 End 3.0 Vehicles: Camera Car, White Car. “...a white car approached from the opposite direction on the left side ... both vehicles attempted to cross the intersection, resulting in a collision between the front-left of the camera car and the front-right of the white car.” R-L 0.50, tIoU 0.	Severity moderate Start 2.0 End 3.0 Vehicles: Camera Car, Red Car . “...several cars were parked, while a few pedestrians were walking ... As the camera car attempted to overtake the black car in front of it, it encountered a red car that had come to a stop ...” Same wrong agent; new unsupported details. R-L 0.31, tIoU 0.25.	Severity moderate Start 2.0 End 3.0 Vehicles: White Car , Red Car. “...a white car attempted to overtake the red car but unexpectedly collided with it. The impact occurred between the front right of the white car and the rear left of the red car.” Fabricated collision, agents altered. R-L 0.31.

TABLE XVII

FULL DECODING ABLATION ON THE FIXED CRASHLOGIC-7B CHECKPOINT ($n=150$). BLEU-4 IS CORPUS-LEVEL. $\dagger$ MARKS A SIGNIFICANT DEGRADATION RELATIVE TO GREEDY ($p < 0.05$). NO VARIANT IMPROVES SIGNIFICANTLY ON ANY COLUMN. MAIN-PAPER SUMMARY: TABLE VII.

Variant	BLEU-4 $\uparrow$	ROUGE-L $\uparrow$	BERTScore $\uparrow$	NLI-Score $\uparrow$	Full-NLI $\uparrow$	$C_H\downarrow$	$C_O\downarrow$
Greedy ($\alpha=0$)	0.142	0.336	0.686	-0.005	0.247	0.223	0.107
TCD $\alpha=0.5$ (reverse)	0.142	0.339	0.686	0.057	0.234	0.212	0.102
TCD $\alpha=1.0$ (reverse)	0.140	0.338	0.684	0.090	0.237	0.227	0.112
TCD $\alpha=1.5$ (reverse)	0.133	0.327	0.681	-0.022	0.223	0.249	0.124 $\dagger$
TCD $\alpha=2.0$ (reverse)	0.133	0.319 $\dagger$	0.676	0.016	0.148 $\dagger$	0.253 $\dagger$	0.128 $\dagger$
TCD $\alpha=0.5$ (shuffle)	0.143	0.340	0.688	-0.025	0.205	0.233	0.114
SEASON (full, $\alpha=1.0$)	0.147	0.332	0.685	0.064	0.204	0.240	0.119

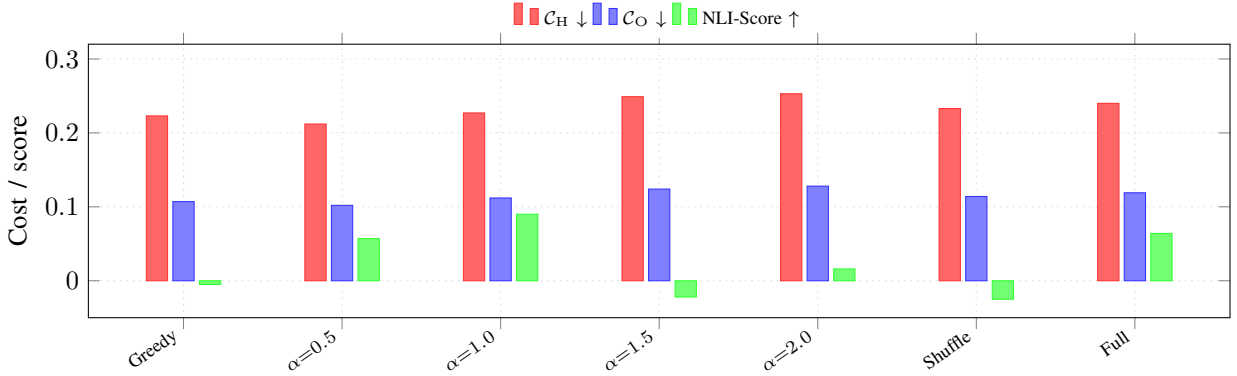


Fig. 3. Decoding ablation on CrashLogic-7B (Table XVII). Hallucination and omission costs are flat from greedy to $\alpha=1.0$ and rise significantly at $\alpha\geq 1.5$; NLI-Score peaks at $\alpha=1.0$ but its paired interval includes zero at every α (Table III, Table V).

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